

SMART BAT – NEGATIVE SCENARIOS & HOW TO HANDLE THEM (BUSINESS VIEW)

I'll break this into **5 big risk categories**, then give **clear solutions** for each.

1 PHYSICAL MISUSE & DAMAGE (BAT / CHIP / CHARGER)

✗ Scenario A: Player intentionally hits ball on chip side / electronics area

Reality:

Yes, some players will misuse equipment.

✓ What YOU do (industry-correct response):

1. Design-level protection (already discussed)

- Electronics **not directly under bat face**
- Offset placement + foam isolation
- Chip not on edge / toe / sweet spot

So even if ball hits “chip side”:

- Shock is absorbed
- Electronics don't take direct impact

2. Policy-level protection (VERY IMPORTANT)

You **explicitly define misuse**.

In warranty terms:

“Damage caused due to intentional impact on electronics housing is not covered.”

This is normal in sports equipment.

Scenario B: Player uses wrong charger / overcharges / cheap adapter

You already handled this correctly by:

- Charging IC (BQ24075)
- Battery with protection PCB

Your stance:

- If charging IC fails → covered
- If battery swells due to misuse → NOT covered after investigation

This is exactly how:

- smartwatches
 - fitness bands
 - wearables operate
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ELECTRONICS FAILURE DURING COACHING (TRUST ISSUE)

Scenario:

Chip stops working during session

OR

Data seems wrong even though player played correct shot

This is **the most dangerous risk** — loss of trust.

✅ **Solution 1: CONFIDENCE LEVEL SYSTEM (CRITICAL)**

Your app should NEVER say:

“Wrong shot.”

Instead it should say:

- ✓ High confidence analysis
- ⚠ Medium confidence
- ✗ Low confidence (data incomplete)

If:

- Camera frame drops
- Sensor desync
- Bluetooth lag

👉 App shows “**Low confidence – not counted**”

This immediately protects trust.

✅ **Solution 2: FAIL SAFE MODE**

If electronics fail mid-session:

- App switches to **camera-only mode**
- OR session is auto-paused

Message shown:

“SmartBat data temporarily unavailable. Session not evaluated.”

This is **far better than wrong feedback**.

CAMERA FAILURE / POOR VIDEO CAPTURE

Scenario:

Phone camera:

- Low light
- Wrong angle
- Poor FPS

Correct response:

The app should:

- Detect camera quality BEFORE session
- Warn:

“Camera angle/lighting insufficient for video analysis.”

Then:

- Allow session
- But **disable shot-selection judgement**

This avoids false negatives.

PLAYER MENTAL IMPACT (VERY IMPORTANT, OFTEN IGNORED)

Scenario:

Player plays well, but app says “bad shot” → confidence drops

This is dangerous.

 YOUR CORE RULE (WRITE THIS DOWN):

**The app never humiliates.
It guides, not judges.**

Instead of:

 “Wrong shot”

Say:

- ✓ “Risky option for this delivery”
- ✓ “Better alternative suggested”
- ✓ “Good intent, timing early”

This keeps players trusting the system.